

Lohitha Saripalli | AI/ML Engineer

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SUMMARY

AI/ML Engineer with 4 years of experience developing and deploying machine learning solutions using Python and SQL across healthcare, banking, and SaaS environments. Skilled in building ML pipelines including data preprocessing, feature engineering, model training, evaluation, and deployment using Scikit-learn, TensorFlow, PyTorch, and XGBoost. Experienced in Generative AI and NLP, developing Retrieval-Augmented Generation (RAG) applications using LangChain, embedding models, semantic search, and LLM workflows. Proficient in Spark, PySpark, ETL pipeline development, FastAPI, Docker, MLflow, and AWS SageMaker for scalable model deployment and monitoring. Strong background in predictive modeling, fraud detection, and healthcare analytics.

SKILLS

Programming & Software Development: Python, SQL, Java, Object-Oriented Programming (OOP), Data Structures & Algorithms, REST APIs, FastAPI, Flask, Git, Linux, Bash

Machine Learning: Supervised & Unsupervised Learning, Classification, Regression, Clustering, Feature Engineering, Feature Selection, Model Training, Hyperparameter Tuning, Model Evaluation, Ensemble Learning, Predictive Modeling, Model Optimization

Deep Learning: TensorFlow, PyTorch, Keras, Neural Networks, CNNs, Transfer Learning, Deep Learning Optimization

Generative AI & LLMs: Retrieval-Augmented Generation (RAG), Large Language Models (LLMs), Prompt Engineering, LangChain, Hugging Face Transformers, Embedding Models, Semantic Search, Vector Search, LLM Application Development, AI Assistants

Natural Language Processing (NLP): Text Classification, Sentiment Analysis, Named Entity Recognition (NER), Information Extraction, Topic Modeling, Text Preprocessing, Tokenization, Clinical Text Analytics

Data Engineering & Big Data: Pandas, NumPy, Apache Spark, PySpark, Hadoop, ETL Pipeline Development, Data Transformation, Data Validation, Workflow Automation, Distributed Data Processing

Databases: PostgreSQL, MySQL, MongoDB, Relational Databases, Data Modeling, Query Optimization

Cloud & MLOps: AWS (S3, SageMaker), Microsoft Azure, Docker, MLflow, CI/CD Pipelines, Model Deployment, Model Monitoring, Experiment Tracking, Containerization

Visualization & Analytics: Tableau, Power BI, Matplotlib, Seaborn, Jupyter Notebook, Data Analysis, Data Visualization

Domain Expertise: Healthcare Analytics, Clinical Data Analytics, Fraud Detection, Risk Analytics, Predictive Analytics

EXPERIENCE

AI/ML Engineer | Cigna Health | OH, USASep 2025 - Present

- Developed predictive machine learning models using Python, Scikit-learn, and XGBoost on healthcare claims datasets, improving high-risk patient identification accuracy by 8% and supporting proactive care management initiatives.
- Engineered scalable feature engineering and ETL pipelines using Pandas, NumPy, Spark, and PySpark, improving data quality and reducing model training time across healthcare analytics workflows.
- Built Retrieval-Augmented Generation (RAG) applications using LangChain, embedding models, and semantic search, reducing manual clinical document review efforts by 25% and accelerating knowledge retrieval.
- Designed NLP pipelines for clinical notes and patient feedback using text classification, sentiment analysis, and information extraction, improving healthcare insight generation and reporting accuracy.
- Trained and optimized deep learning models using TensorFlow and PyTorch for clinical text analytics, enhancing recommendation quality and improving patient engagement initiatives.
- Deployed scalable machine learning inference services using FastAPI, Docker, MLflow, and AWS SageMaker, enabling model versioning, experiment tracking, and reliable production deployments.
- Collaborated with data engineers, analysts, and healthcare stakeholders to integrate PostgreSQL and MySQL data sources into ML workflows, improving accessibility and consistency of enterprise healthcare data.

AI/ML Engineer | HCL | IndiaJun 2022 - Dec 2023

- Developed fraud detection models using supervised and unsupervised machine learning techniques, improving anomaly detection accuracy by 18% across high-volume banking transaction datasets.
- Designed distributed data processing pipelines using Spark, PySpark, and Hadoop to transform large-scale financial datasets for fraud detection and credit risk modeling applications.
- Built credit risk prediction models using Scikit-learn, TensorFlow, and ensemble learning techniques, improving loan default prediction accuracy by 15% and supporting underwriting decisions.
- Implemented NLP solutions for financial documents and customer communications using text classification and information extraction, improving compliance monitoring and document processing efficiency.
- Optimized machine learning training workflows using Python, Pandas, and NumPy, reducing processing time and improving model development efficiency across multiple projects.
- Deployed containerized machine learning applications using Docker and CI/CD pipelines, enabling reliable model delivery and seamless integration into enterprise banking platforms.

ML Engineer | Freshworks | IndiaJan 2021 - May 2022

- Developed customer churn prediction models using Python, Scikit-learn, and ensemble learning algorithms, improving prediction accuracy by 16% and supporting targeted customer retention strategies.
- Engineered ETL and data transformation pipelines to process large-scale product usage data, improving data quality and accelerating machine learning workflow execution.
- Built recommendation systems using machine learning and deep learning techniques, delivering personalized user experiences that increased customer engagement and feature adoption.
- Implemented NLP-based automation solutions for customer support conversations using text classification and sentiment analysis, reducing manual ticket triage efforts and improving operational efficiency.
- Developed RESTful prediction APIs using FastAPI and Docker, enabling scalable deployment of machine learning models across SaaS products and internal business applications.
- Partnered with analytics and product teams to monitor model performance, generate reporting datasets, and deliver actionable insights into customer behavior and product usage.

EDUCATION

Master's in Computer Science | University of Toledo | OH, USADec 2025

PROJECTS

AI Healthcare Assistant (NLP, Scikit-learn, Django, REST APIs)

- Developed an NLP-based healthcare assistant that analyzes patient-reported symptoms and provides preliminary condition suggestions using Scikit-learn and classical ML models.
- Built a Django-powered chatbot with REST APIs to recommend relevant medical specialties and guidance, integrating text-processing pipelines for automated symptom interpretation and improved patient triage support.

Gesture-Based Media Control (TensorFlow, Keras, OpenCV)

- Built a real-time hand-gesture recognition system using CNNs and transfer learning (ResNet/VGG) to control media playback actions such as play, pause, volume, and skip.
- Implemented webcam-based gesture detection with OpenCV and optimized the model for low-latency inference, enabling smooth interaction for smart TVs, kiosks, and accessibility applications.

ACHIEVEMENT

- Gender-Specific Osteoporosis Risk Prediction Using Longitudinal Clinical Data and Machine Learning | medRxiv ([Link](#))

CERTIFICATIONS

- Python for Everybody Specialization
- Python Foundation with JavaScript, HTML, and CSS
- AWS Certified AI Practitioner